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Bunion correction system and method

Abstract

A minimally invasive method of correcting a bunion includes performing an osteotomy to divide the metatarsal into first and second portions, creating a pocket in the first portion implanting a nail in the pocket, securing a suture to the joint capsule, tensioning the suture to align the great toe with the metatarsal, attaching the suture to the nail, and fastening the nail to the second portion. The nail includes an anchor portion for anchoring in the first portion, a head for attachment to a second portion, a passage for attachment of the suture, a first aperture for a fastener to attach the head with the first portion, and a second aperture for a fastener to attach the head with the second portion.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY RELATED APPLICATIONS (1) The is a national phase application of PCT Application No. PCT/US2020/045210, filed Aug. 6, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/883,819, titled “BUNION CORRECTION SYSTEM AND METHOD,” filed Aug. 7, 2019, the entirety of which is hereby incorporated by reference.

FIELD

(1) The present invention relates to surgical treatment for deformities of the foot. More specifically, the present invention relates to implants, instrumentation and methods for minimally invasive bunion correction.

BACKGROUND

(2) Bunions are a progressive disorder typically beginning with a leaning of the great toe, which may gradually change the angle of the bones and produce a characteristic bump on the medial side of the metatarsal near the joint of the metatarsal with the proximal phalanx. Specifically, the bunion is the prominence made of bone and at times an inflamed bursa. Hallux valgus is the condition in which the great toe deviates from the normal position toward the direction of the second toe.

(3) Bunion correction or repair is a common surgery with over 100,000 surgeries performed annually in the US. Many surgical procedures for bunion repair are invasive and painful, requiring an incision of several inches and a long period of convalescence, of up to 10-12 weeks. Minimally invasive surgery has been performed in orthopedics for decades. However, creating the bone cuts

has been performed with burrs and drill bits inserted blindly through small incisions. This method of surgery lends itself to potential adjacent soft tissue damage and unrepeatable results from patient to patient. The disclosure contained herein seeks to remedy this problem by providing an instrumented technique and guides to provide repeatability and limit the damage to tissue along with a simple implant insertion technique.

(4) Disclosed herein is an implant and method for bunion repair which can be performed as a minimally invasive procedure, thus reducing discomfort, scarring and recovery time in comparison with more invasive bunion correction procedures.

SUMMARY OF THE INVENTION

(5) The various systems and methods of the present invention have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available techniques. The systems and methods of the present invention may provide techniques for bunion correction which result in a streamlined procedure, faster recovery, reduced scarring, and reduced discomfort during healing.

(6) To achieve the foregoing, and in accordance with the invention as embodied and broadly described herein, one aspect of the disclosure is a first method for correcting a bunion formed at the joint between a metatarsal and a great toe. An incision is made along a side of the metatarsal. A target location is selected on the metatarsal. The metatarsal is resected into a first metatarsal portion and a separate second metatarsal portion at the target location. The first metatarsal portion has a distal-facing surface created by the resecting. A pocket instrument is inserted into the first metatarsal portion at the distal-facing surface to create a pocket in the first metatarsal portion at the distal-facing surface. An implant is inserted through the incision into the pocket of the first metatarsal portion, the implant having a monolithic body with a head and an anchor.

(7) In another aspect of the first method, the pocket extends into an intramedullary canal of the first metatarsal portion through the distal-facing surface.

(8) In another aspect of the first method, at least one k-wire inserts into the first metatarsal portion through the distal-facing surface. The first k-wire guides the first instrument into the first metatarsal portion to create the pocket.

(9) In another aspect of the first method, the first pocket instrument is a broach.

(10) In another aspect of the first method, the broach includes a handle aligned along a first axis, a insertion portion having a plurality of teeth aligned along a second axis, and an offset portion between the handle and the insertion portion such that the first axis is angled with respect to the second axis.

(11) In another aspect of the first method, the implant head attaches to the first metatarsal portion.

(12) In another aspect of the first method, the implant head attached to the first metatarsal portion includes inserting a screw through an aperture of the implant head and into the first metatarsal portion.

(13) In another aspect of the first method, the implant head attaches to the second metatarsal portion.

(14) In another aspect of the first method, a length of suture is secured to the great toe, tensioning the suture to re-align the great toe relative to the first metatarsal portion, and attaching the length of suture to the implant head.

(15) In another aspect of the first method, the second metatarsal portion translates to expose the distal-facing surface on the first metatarsal portion.

(16) Another aspect of the disclosure is a second method for correcting a bunion. An incision is made along a side of the metatarsal. A target location is selected on the metatarsal. The metatarsal is resected into a first metatarsal portion and a separate second metatarsal portion, the first metatarsal portion having a distal-facing surface created by the resecting. An implant is implanted through the incision into the first metatarsal portion. The implant has a monolithic body having a head and an anchor, the anchor extending along an implant axis. The implant head is firstly

attached to the first metatarsal portion at the distal-facing surface; and the implant head is secondly attached to the second metatarsal portion.

(17) In another aspect of the second method, the implant head is attached to the first metatarsal portion and includes inserting a first screw through a first aperture of the implant head and into the distal-facing surface of the first metatarsal portion.

(18) In another aspect of the second method, the first aperture is aligned along a first axis at a first angle relative to the implant axis.

(19) In another aspect of the second method, the first angle is less than approximately 45°.

(20) In another aspect of the second method, attaching the implant head to the second metatarsal portion includes inserting a second screw through a second aperture of the implant head and into the second metatarsal portion.

(21) In another aspect of the second method, the second aperture is aligned along a second axis at a second angle relative to the implant axis, the second angle being greater than the first angle.

(22) In another aspect of the second method, the second angle is greater than 60°.

(23) In another aspect of the second method, the second metatarsal portion translates to expose the distal-facing surface on the first metatarsal portion. A pocket is created in the first metatarsal portion at the distal-facing surface, and the pocket extends into an intramedullary canal of the first metatarsal portion through the distal-facing surface.

(24) In another aspect of the second method, a pocket instrument is inserted and guided by at least one k-wire into the first metatarsal portion at the distal-facing surface to create the pocket.

(25) In another aspect of the second method, a length of suture is secured to the great toe, tensioning the suture to re-align the great toe relative to the first metatarsal portion, and attaching the length of suture to the implant.

(26) Another aspect of the disclosure is a third method for correcting a bunion. An incision is made along a side of the metatarsal. A first k-wire is introduced through the incision and into the metatarsal at a selected target location. The metatarsal is resected into a first metatarsal portion and a separate second metatarsal portion at the selected target location. The first metatarsal portion has a distal-facing surface created by the resecting. A second k-wire is inserted into the first metatarsal portion at the distal-facing surface. A pocket instrument is inserted into the first metatarsal portion at the distal-facing surface guided by the second k-wire to create a pocket in the first metatarsal portion at the distal-facing surface. An implant is inserted through the incision into the first metatarsal portion, the implant having a monolithic body having a head and an anchor, the anchor extending along an implant axis. The implant head is attached to the first metatarsal portion at the distal-facing surface and attaching the implant head to the second metatarsal portion.

(27) In another aspect of the third method, attaching the implant head to the first metatarsal portion includes inserting a first screw through a first aperture of the implant head and into the distal-facing surface of the first metatarsal portion and attaching the implant head to the second metatarsal portion includes inserting a second screw through a second aperture of the implant head and into the second metatarsal portion.

(28) In another aspect of the third method, the second metatarsal portion translates to expose the distal-facing surface on the first metatarsal portion; and a pocket is created in the first metatarsal portion at the distal-facing surface, wherein the pocket extends into an intramedullary canal of the first metatarsal portion through the distal-facing surface.

(29) These and other features and advantages of the present invention will become more fully apparent from the following description and appended claims, or may be learned by the practice of the invention as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the invention will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings.

Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the invention's scope, the exemplary embodiments of the invention will be described with additional specificity and detail through use of the accompanying drawings in which:

- (2) FIG. 1A is a perspective view of a bunion correction implant, comprising a nail and a fastener, according to one embodiment of the invention; FIG. 1B is an exploded view of the implant;
- (3) FIG. 2 is a perspective view of a partial skeleton of a foot, with the implant of FIG. 1A implanted into the first metatarsal;
- (4) FIG. 3A is medial view of the nail of FIG. 1A; FIG. 3B is lateral view of the nail of FIG. 1A; FIG. 3C is a superior view of the nail of FIG. 1A;
- (5) FIG. 4 is a perspective view of the foot skeleton with a k-wire inserted into the metatarsal and a first guide mounted on the k-wire, the first guide having a plurality of guide holes;
- (6) FIG. 5 is a perspective view of the foot skeleton, k-wire and guide of FIG. 4, with additional k-wires inserted into the metatarsal;
- (7) FIG. 6 is a perspective view of the foot skeleton, k-wires and first guide of FIG. 5, with a trocar extending through one of the guide holes to create a hole in the metatarsal;
- (8) FIG. 7 is a perspective view of the foot skeleton and k-wires of FIG. 5, with a second guide mounted on the k-wires;
- (9) FIG. 8 is a perspective view of the foot skeleton, k-wires and second guide of FIG. 7, with a broach extending through a guide slot to cut an osteotomy in the metatarsal and separate the metatarsal into a proximal metatarsal portion and a distal metatarsal portion;
- (10) FIG. 9 is a perspective view of the foot skeleton of FIG. 8, with the distal metatarsal portion shifted laterally relative to the proximal metatarsal portion;
- (11) FIG. 10 is a perspective view of the foot skeleton of FIG. 9, with the nail of FIG. 1 mounted on an implant inserter and implanted into the proximal metatarsal portion;
- (12) FIG. 11A is a medial view of the foot with the nail of FIG. 1 implanted in the proximal metatarsal portion and a needle and suture inserted into the foot through an incision at the site of the implant and exiting at a first location on the great toe at the proximal phalanx; FIG. 11B is a medial view of the foot of FIG. 11A, with a first stitch being made at the first location on the great toe; FIG. 11C is a medial view of the foot of FIG. 11B, with the needle emerging at a second location on the great toe; FIG. 11D is a medial view of the foot of FIG. 11C, with a second stitch being made at the second location on the great toe; FIG. 11E is a medial view of the foot of FIG. 11D, with the suture routed through an implant bore, and the needle and suture emerging through the incision, and showing a path of the suture;
- (13) FIG. 12 is a perspective view of the foot skeleton of FIG. 10 showing the suture and the fastener aimed toward an opening of the nail;
- (14) FIG. 13A is a perspective view of the first guide of FIG. 5; FIG. 13B is a medial view of the first guide of FIG. 5; FIG. 13C is a lateral view of the first guide of FIG. 5; FIG. 13D is a superior view of the first guide of FIG. 5; FIG. 13E is a distal view of the distal end of the metatarsal, the first guide, and the trocar of FIG. 6 extending through another one of the guide holes, with dotted lines indicating the trajectories of the plurality of guide holes;
- (15) FIG. 14A is a medial perspective view of the second guide of FIG. 7;
- (16) FIG. 14B is a medial view of the second guide of FIG. 7;
- (17) FIG. 14C is a side view of the second guide of FIG. 7;
- (18) FIG. 14D is a lateral perspective view of the second guide of FIG. 7;
- (19) FIG. 14E is a medial view of another embodiment of a second guide;

- (20) FIG. 15 is a perspective view of the broach of FIG. 8;
- (21) FIG. 16 is a perspective view of the implant inserter of FIG. 10;
- (22) FIG. 17 is a perspective view of the foot skeleton of FIG. 9, showing the insertion of a k-wire into the proximal metatarsal portion;
- (23) FIG. 18 is a perspective view of the foot skeleton of FIG. 17, showing a pocket instrument for forming a pocket inserted into the proximal metatarsal portion as guided by the k-wire;
- (24) FIG. 19 is a side view of the pocket instrument shown in FIG. 18;
- (25) FIG. 20 is a bottom view of the pocket instrument shown in FIG. 18;
- (26) FIG. 21 is a perspective view of another bunion correction implant;
- (27) FIG. 22 is an exploded view of the implant of FIG. 21;
- (28) FIG. 23 is a medial view of a nail of the implant of FIG. 21;
- (29) FIG. 24 is a lateral view of the nail of FIG. 23;
- (30) FIG. 25 is a section view of the implant of FIG. 21;
- (31) FIG. 26 is a medial view of a nail of another implant system;
- (32) FIG. 27 is a lateral view of the nail of FIG. 26;
- (33) FIG. 28 is a superior view of the nail of FIG. 26;
- (34) FIG. 29 is a section view of the implant of FIG. 26.

DETAILED DESCRIPTION

(35) Exemplary embodiments of the invention will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the invention, as generally described and illustrated in the Figures herein, could be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the apparatus, system, and method, as represented in FIGS. 1A through 25, is not intended to limit the scope of the invention, as claimed, but is merely representative exemplary of exemplary embodiments of the invention.

(36) The phrases “connected to,” “coupled to” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be functionally coupled to each other even though they are not in direct contact with each other. The term “abutting” refers to items that are in direct physical contact with each other, although the items may not necessarily be attached together. The phrase “fluid communication” refers to two features that are connected such that a fluid within one feature is able to pass into the other feature.

(37) Directional and/or relational terms such as, but not limited to, left, right, superior, inferior, top, bottom, vertical, horizontal, medial, and lateral are relative to each other, are dependent on the specific orientation of an applicable element or article, are used accordingly to aid in the description of the various embodiments in this specification and the appended claims, and are not necessarily intended to be construed as limiting. Standard medical terminology may be used to describe human anatomy, or the relationship of objects to the human anatomy. For example, proximal refers to an object or anatomical element closer to the center of the body, while distal refers to an object or anatomical element farther away from the center of the body.

(38) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

(39) Referring to FIG. 1A, a bunion correction implant system **100** according to one embodiment of the invention includes a nail **102** and a fastener **104**. As seen in FIG. 2, the nail **102** may be implanted to extend longitudinally into a proximal portion **2a** of a resected metatarsal **2**, and the fastener **104** inserted through a portion of the nail to secure it to the distal portion **2b** of the metatarsal **2**. The bunion correction implant system **100** may further include a suture which may be

routed through the medial capsule of the metatarsophalangeal (MTP) joint, secured in the soft tissues of the great toe, tensioned to re-align the position of the proximal phalanx **4** relative to the metatarsal, and secured to the nail.

(40) Referring to FIGS. **1A-B**, and **3A-C**, the implant nail **102** is a monolithic body extending from a first end **106** which may be a distal end, to a second end **108** which may be a proximal end. The nail **102** may be generally rectangular in cross-section, having a medial side **110** which may be an outer side, a lateral side **112** which may be an inner side, a superior side **114**, and an inferior side **116**. The nail **102** includes a head **120**, an anchor **122**, and a neck **124** extending between the head **120** and the anchor **122**.

(41) With particular reference to FIGS. **3B** and **3C**, the nail anchor **122** and neck **124** extend along a longitudinal axis **105**, and the nail head **120** extends distally away from the neck **124** at an angle. The nail head extends between a head first end **126** and a head second end **128** along a head first axis **107**. In the embodiment depicted, the angle α between the head first axis **107** and the longitudinal axis **105** is 15° , with an angle β between the neck lateral surface and the head lateral surface of 165° . In other embodiments of the invention, angle α may be in the range of 0° to 25° . In at least the embodiment depicted, the nail **102** and assembled implant **100** are bilaterally symmetrical with respect to the longitudinal axis **105**, and with respect to the head first axis **107**.

(42) The nail head **120** includes an opening **130** centered on a head second axis **109**, which is perpendicular to the head first axis **107**. The head second axis may extend generally medially-laterally (ML) upon implantation. In the embodiment depicted, the opening **130** extends between a head lateral side **115** and a head medial side **113**, and includes threads **132** for engagement with the fastener **104**, although additional embodiments may lack threads. A concave lip **134** encircles the opening **130**. The head second end **128** is wider than the neck **124** with respect to the superior-inferior dimension, and includes a first shoulder **140** and a second shoulder **142**, the shoulders **140**, **142** projecting superiorly and inferiorly, respectively, away from the neck **124** at the intersection of the neck and the head. The first shoulder **140** includes a first proximal shoulder surface **141**, and the second shoulder **142** includes a second proximal shoulder surface **143**. The proximal shoulder surfaces **141**, **143** face proximally away from the head **120** and are at right angles to the neck superior and inferior sides **114**, **116**. The head **120** may further include a transverse bore **148**, extending along a head third axis **111** which is perpendicular to the head first axis **107** and the head second axis **109**. The head third axis **111** may extend generally superiorly-inferiorly (SI) upon implantation. In the embodiment depicted, the thickness of the head **120** between its medial **113** and lateral **115** sides increases between the head first end **126** and the head second end **128**, so that the thickest part of the head is at the shoulders **140**, **142**.

(43) The neck **124** extends between and connects the head **120** with the anchor **122**. The thickness of the neck **124** between the medial **110** and lateral **112** sides can vary depending on the desired degree of shift of the metatarsal. In at least the embodiment depicted, the neck thickness tapers between the head **120** and the anchor **122**. The width of the neck **124** between the superior **114** and inferior **116** sides may also vary. The length of the nail between the first and second ends **106**, **108** can vary, as can the relative lengths of the head, neck, and/or anchor portions. The anchor **122** is coaxial with the neck **124**, and extends from the neck to the second end **108** of the nail. Both the thickness of the anchor between the medial **110** and lateral **112** sides, and the width of the anchor between the superior **114** and inferior **116** sides may taper towards the nail second end, promoting easy insertion of the nail into bone. The anchor second end may be rounded as in the embodiment depicted in FIGS. **1A** and **1B**; in other embodiments it may be pointed, flattened, serrated, or another shape. The anchor **122** includes a plurality of bone engagement features **144** which may be shaped as teeth, scallops, serrations, or other shapes to promote engagement within bone. For example, the scallops **146** in the embodiment depicted provide surface irregularities which resist nail backup. In the embodiment shown, the neck and anchor are free from openings; other embodiments could include openings for supplementary fixation or instrument connection.

(44) Fastener **104** includes a fastener head **150**, fastener shaft **152**, and tip **154**. The head **150** includes threads **156** for locking engagement with threads **132** in the nail head **120**; other embodiments may lack threads **156**. The shaft **152** includes threads **158** for engagement in bone. The head **150** may include a driving feature **157** for engagement with a driver. In the embodiment depicted, fastener **104** is a locking screw type fastener; in other embodiments the fastener may be locking or non-locking, and may be polyaxially adjustable or non-polyaxially adjustable.

(45) The nail **102** and fastener **104** may comprise titanium, stainless steel, polyether ether ketone (PEEK), nitinol, and/or other rigid biocompatible materials or combinations thereof. The suture is a non-resorbable suture, although other embodiments may include a resorbable suture.

(46) Referring to FIGS. **4-16**, a method of correcting a bunion includes one or more of the following steps. Although the steps are described in an order, in other embodiments of the method one or more of the steps may be repeated, omitted, or performed in a different order.

(47) A small incision is made in the affected metatarsal at the medial side of the bunion. Preferably, the incision is 0.5 inch long or less. Referring to FIG. **4**, a first k-wire **159** is introduced through the incision and into the metatarsal at a selected target location. A first guide block **164** is mounted onto the k-wire **159** and urged toward the metatarsal **2**. Referring to FIG. **5**, second and third k-wires **160**, **162** are introduced through the guide block **164** into the metatarsal, on either side of the selected target location. The second k-wire **160** is located in a proximal metatarsal portion **2a**, and the third k-wire **162** is introduced into a distal metatarsal portion **2b**. The first k-wire **159** is removed from the metatarsal.

(48) As shown in FIGS. **13A-E**, the first guide block **164** extends between a first or lateral side **166**, and a second or medial side **168**. First and second guide portions **170**, **172** project superiorly and inferiorly, respectively. A series of guide holes **174** extend through the guide block, each defining a trajectory **175**, which converge at a common point, as seen in FIG. **13E**. The guide holes **174** and their trajectories **175** are coplanar, defining a cut plane when the first guide block **164** is mounted on the k-wires **160**, **162**. A pair of mounting holes **176**, **178** are sized to slide over the k-wires. First and second mounting supports **180**, **182** extend medially from the guide block **164** and are separated by a gap **184**. The mounting supports **180**, **182** include slots **186**, **188** for guiding and supporting the k-wires, and prevent rotation of the guide block **164** once mounted. The medial side **168** of the guide block **164** may be convexly curved as shown.

(49) Referring to FIGS. **6** and **13E**, after the first guide block **164** is mounted on the k-wires and abuts the metatarsal **2**, a trocar **192**, reamer or other instrument is introduced through one of the guide holes **174** and inserted into the metatarsal **2** to create a hole through the metatarsal along the trajectory of the guide hole. The trocar **192** may then be inserted through more of the guide holes **174** and through the metatarsal. This step creates a linear series of openings through the bone at the targeted location, weakening the bone at that location in preparation for a subsequent osteotomy to resect the metatarsal head. After the desired number of openings are created, the first guide block **164** is withdrawn from the k-wires.

(50) Referring to FIG. **7**, a second guide block **200** is introduced onto the k-wires **160**, **162**. As shown in FIGS. **14A-D**, the second guide block **200** extends between a first or lateral side **202**, and a second or medial side **204**. First and second guide portions **206**, **208** project superiorly and inferiorly, respectively. A cutting slot **210** extends through the guide block **200**, opening out on the lateral and medial sides **202**, **204**, and mounting holes **212**, **214** extend between the medial and lateral sides for mounting the block on the k-wires. The guide block **200** may be curved as shown, wherein the medial side **204** of the guide block **200** is convexly curved and the lateral side **202** is concave. At least the concave lateral surface may allow the guide to fit closely against the targeted location. When the second guide block **200** is mounted on the k-wires **160**, **162**, the mid-plane of the opening defined by the cutting slot **210** is coplanar or at least parallel with the cut plane defined by the first guide block guide holes **174** and trajectories **175**, and with the series of openings created in the bone in the previous step.

(51) FIG. 14E shows an alternative embodiment of a second guide block **200a**, similar to the second guide block **200** with the differences noted below. The second guide block **200a** can extend between a first or medial side (**204a**) and a lateral side (not shown). The lateral side can be concave and the medial side **204a** can be convex. A cutting slot **210a** extends through the guide block **200a**. The cutting slot **210a** can include a first concave channel **210b** on a first side of the cutting slot **210a** and/or a second concave channel **210c** on a second side of the cutting slot **210a**. The channels **210b**, **210c** can form an aperture for receiving a k-wire (e.g., first k-wire **159**). First and second guide portions **206a**, **208a** project superiorly and inferiorly from the cutting slot **210a**. Mounting holes **212a**, **214a** extend between the medial and lateral sides. The second guide block **200a** can be mounted first on a k-wire through the channels **210b**, **210c** and urged toward the metatarsal 2 (See FIG. 4). Second and third k-wires **160**, **162** can then be introduced through the mounting holes **212a**, **214a** and into the metatarsal 2. The first k-wire **159** can then be removed from the metatarsal 2.

(52) Referring to FIGS. 8 and 15, a broach **220** is used to create an osteotomy in the metatarsal 2, resecting the metatarsal into the first or proximal metatarsal portion **2a** and the second or distal metatarsal portion **2b**. The broach **220** includes a handle portion **222**, a shaft portion **224**, and an insertion portion **226** with a cutting tip **228** and cutting edges **230**, **232**. The cutting edges and tip may be beveled, sharpened, serrated and/or otherwise configured to cut through bone. The broach insertion portion **226** is urged laterally through the cutting slot **210** and into the bone to create the osteotomy. The shaft portion **224** may act as a stop to limit lateral insertion of the insertion portion through the cutting slot. In other embodiments of the method, instead of or in combination with broach **220**, a saw, blade, chisel, osteotome, curette, pick, rasp or other instrument or combinations thereof may be used to perform the osteotomy. When the osteotomy step is completed, the second guide block **200** is removed from the k-wires. Similarly, a cutting instrument, such as those listed above, can be urged laterally through the cutting slot **210a** of the second guide block **200a** and into the bone to create the osteotomy.

(53) Referring to FIG. 9, the now separate distal metatarsal bone portion **2b** is translated laterally relative to proximal metatarsal bone portion **2a**. A generally flat distal-facing surface **2c** on the proximal metatarsal portion **2a** is exposed, and it is into this surface that the nail **102** is implanted. Distal-facing surface **2c** of the proximal metatarsal bone portion **2a**, and a medial-facing surface **2d** of the distal metatarsal bone portion **2b** may be referred to as abutment surfaces. The degree of offset of the distal metatarsal portion may vary but is sufficient to permit implantation of nail **102** into the distal-facing surface **2c** so that the nail head **120** does not protrude medially beyond the medial outer surface of the proximal metatarsal portion **2a** after implantation. The k-wires **160**, **162** may be removed before or after the shifting of the distal metatarsal portion relative to proximal metatarsal portion.

(54) Referring to FIGS. 10 and 16, the nail **102** is inserted through the incision and anchored into the prepared proximal metatarsal portion **2a**. Prior to implantation, a suture **250** may be introduced to extend through the transverse bore **148**. An implant inserter such as inserter **240** may be employed to implant the nail **102** into the bone. Inserter **240** comprises a handle portion **242**, a shaft portion **244**, and an implant engagement end **246**. Threads **248** are formed on the implant engagement end **246**, which may cooperate with the nail threads **132** to removably attach the nail **102** to the inserter **240**. The inserter **240** is moved to insert the anchor **122** and neck **124** proximally into the distal-facing surface **2c** and into the intramedullary canal of the metatarsal, leaving the head **120** distal to the proximal metatarsal portion **2a**. If needed, the inserter **240** may be tapped to drive the nail **102** into position in the proximal metatarsal portion **2a**. The nail **102** is positioned so that the proximal surfaces **141**, **143** of shoulders **140**, **142** abut the prepared distal-facing surface **2c** of the metatarsal, and the nail head lateral side **115** is immediately adjacent to medial surface **2d** of the distal metatarsal portion **2b**. When the nail **102** is properly seated in the desired location, the inserter **240** may be rotated to disengage it from the implanted nail **102**.

(55) Referring to FIGS. 11A-11F and 12, the suture 250 may be engaged with implant 100 and secured to soft tissues of the great toe 6 to change the alignment of the phalanx 4 relative to the metatarsal 2 and correcting hallux valgus. As shown in FIG. 11A, a needle 260 carrying suture 250 is introduced through the incision 5, enters the medial capsule and emerges at a first location 6a on the epidermis of the great toe. The suture includes a first end 252 and a second end 254. As shown in FIG. 11B, the needle re-enters the skin at location 6a, creating a first stitch 256 in the soft tissue surrounding the phalanx 4. Continuing to FIG. 11C, the needle 260 and suture 250 emerge at a second location 6b on the epidermis of the great toe. Referring to FIG. 11D, the needle 260 and suture 250 re-enter the great toe at location 6b, creating a second stitch 258 in the soft tissue of the great toe 6. As shown in FIG. 11E, the needle and suture second end 252 emerge through the incision 5. The suture 250 is tensioned to change the alignment of the phalanx 4 relative to the metatarsal 2, providing tensile force along the medial side of the phalanx and correcting the hallux valgus. The nail shoulders 140, 142, abutting against the proximal metatarsal portion 2a act as a buttress to support the tension and alignment correction. The tensioned suture 250 is attached to the nail 102, with one or both of the first and second ends 252, 254 passing through the transverse bore 148. The suture first end 252 may pass through the transverse bore 148 from the inferior side 116 to the superior side 114 of the nail 102, and a knot 262 may be tied at the superior side of the bore 148, maintaining the suture tension and the correction. The knot 262 may be wider than the diameter of the transverse bore 148, so that the knot cannot pass through the bore. After knotting, the remaining suture free ends 252, 254 may be trimmed off. As seen in FIGS. 11E and 12, the suture 250 may follow a three-sided path from the implant 100 to the first and second stitches in the great toe, and back to the implant 100.

(56) Referring to FIGS. 12 and 2, the screw tip 154 and shaft 152 are inserted through the nail opening 130 to secure the nail head 120 to the distal metatarsal portion 2b. As the shaft threads 158 engage in the bone, the nail head 120 lateral side 115 is urged to medial surface 2d of the distal metatarsal portion 2b. The screw head threads 156 engage with the nail opening threads 132 to lock the screw 104 to the nail 102. The incision 5 is closed. Following closure of the incision, the suture 250 remains secured to the great toe and attached to the implant 100.

(57) FIGS. 17-18 show an optional additional step in the insertion process for the bunion correction implant system 100 into a patient's foot, as shown and described in FIGS. 1-16 and the accompanying text. FIG. 17 shows the separate distal metatarsal portion 2b and the proximal metatarsal portion 2a before the insertion of the nail 102 into the proximal metatarsal portion 2a. The distal metatarsal portion 2b can include the medial facing surface 2d. The proximal metatarsal portion 2a can include the distal facing surface 2c. A k-wire 360 can be inserted into the proximal metatarsal portion 2a. The k-wire 360 can be inserted through the distal facing surface 2c. The k-wire 360 can include an end 362. The end 362 can include a sharpened point. The end 362 can be inserted into the intramedullary portion of the metatarsal 2. The k-wire 360 can be inserted into the proximal metatarsal portion 2a after the distal metatarsal portion 2b is shifted relative to the proximal metatarsal portion 2a (e.g., to expose the distal facing surface 2c).

(58) The k-wire 360 can be inserted at an angle corresponding to the desired positioning of the implant 100 within the proximal metatarsal portion 2a. The entry point and/or the angle of entry of the k-wire 360 into the distal facing surface 2c can determine the amount of offset between the distal metatarsal portion 2b and the proximal metatarsal portion 2a. The k-wire 360 can accordingly be used to estimate the final positioning of the distal metatarsal portion 2b and the proximal metatarsal portion 2a. If the k-wire 360 is inserted in an undesirable position, the k-wire can be easily removed and repositioned within the proximal metatarsal portion 2a. The k-wire 360 can be used as a guide for a pocket instrument 320. As shown in FIG. 18, the pocket instrument 320 can be used to remove bone material from the proximal metatarsal portion 2a to form a pocket 380 for the nail 102. The nail 102 of the implant 100 can be inserted into the proximal metatarsal portion 2a (e.g., into the pocket 380) using the steps and/or tools described above in FIG. 10 and

the accompanying text. The nail head **120** can be aligned and attached with the medially facing surface **2d** of the distal metatarsal portion **2b**.

(59) As shown further in FIGS. **19-20**, the pocket instrument **320** can include a handle **322**. The pocket instrument **320** can include an insertion portion **326**. The handle **322** can be generally aligned along a first axis **323**. The insertion portion **326** can be generally aligned along a second axis **327**. In one implementation, the first and second axes **323**, **327** are parallel, although this is not required. The insertion portion **326** can be connected at the handle **322** by an offset neck **324**. The offset neck **324** can space the handle portion **322** apart from the insertion portion **326**.

(60) The insertion portion **326** can be a broach, punch, or a non-rotating cutting instrument. The insertion portion **326** can include one or more cutting edges **330**. The cutting edges **330** can be used to remove material from the proximal metatarsal portion **2a** to form the pocket **380**. Alternatively, the insertion portion **326** can be smoothed. The insertion portion **326** can include a tip **328**. The insertion portion **326**, whether smoothed or including the cutting edges **330**, can be used to create a trial pocket for the nail **102**. The shape of the pocket **380** can correspond to the shape of the nail **102** of the implant **100** (e.g., the insertion portion **326**). The insertion portion **326** can include a lateral side **312** and a medial side **310**. The insertion portion **326** can include a superior side **314** and an inferior side **316**. The insertion portion **326** can be non-cylindrical in cross sectional shape (e.g., rectangular, as shown). The non-cylindrical in cross sectional shape can enhance stability of the nail **102** within the pocket **380**. The pocket **380** (and/or the insertion portion **326**) can be smaller, in one or more dimensions and/or along one or more sides, than the nail **102**. The nail **102** can be in a press-fit condition within the pocket **380** (e.g., engaged with one or more opposing sides of the pocket **380**) after insertion. In some implementations, the aperture **380** and insertion portion **326** can be sized to the dimensions of the nail **102**.

(61) The insertion portion **326** can include a channel **340**. The channel **340** can extend along the second axis **327** of the insertion portion **326**. The channel **340** can include a proximate opening **342** and a distal opening **344**. The proximal opening **342** can be adjacent to the offset neck portion **324**. The distal opening **344** can be adjacent to the tip **328**.

(62) In use to form the pocket **380**, the insertion portion **326** of the pocket instrument **320** can be guided into the proximal metatarsal portion **2a** along the k-wire **360**. The offset neck **324** can space the handle portion **322** from the k-wire **360** to improve the ergonomics of the pocket instrument **320**. The k-wire **360** can be received within the channel **340**. The pocket instrument **320** can be slid along a length of the k-wire **360** into the proximal metatarsal bone portion **2a**. The pocket instrument **320** can be moved in one or more sawing strokes or impacts (e.g., from a hammer striking the pocket instrument **320**) to form the pocket **380**. After forming the pocket **380**, the k-wire **360** and the insertion portion **326** can be removed from the proximal metatarsal portion **2a**. This implant system **100** can be installed within the pocket **380**, as described above.

(63) Referring to FIGS. **21-25**, a bunion correction implant system **400** according to another embodiment of the invention includes a nail **402**, a first fastener **404a**, and a second fastener **404b**. The nail **402** may be implanted to extend longitudinally into the proximal metatarsal portion **2a** of the resected metatarsal **2**. The first fastener **404a** can be inserted through a first portion of the nail **402** to secure it to the distal metatarsal portion **2b** of the metatarsal **2**. The second fastener **404b** can be inserted through a second portion of the nail **402** to secure it to the proximal metatarsal portion **2a** of the metatarsal **2**. The implant system **400** may further include a suture which may be routed through the medial capsule of the MTP joint, secured in the soft tissues of the great toe, tensioned to re-align the position of the proximal phalanx **4** relative to the metatarsal, and/or secured to the nail, as described above in relation to FIGS. **1-20** and the accompanying description.

(64) The implant nail **402** can be a monolithic body extending from a first end **406** which may be a distal end, to a second end **408** which may be a proximal end. The nail **402** may be generally rectangular in cross-section. The nail **402** can have a medial side **410** which may be an outer side, a lateral side **412** which may be an inner side, a superior side **414**, and/or an inferior side **416**. The

nail **402** can include a head **420**. The head **420** can be one the first end **406**. The nail **402** can include an anchor **422**. The anchor **422** can be on the second end **408**. The nail **402** can include a neck **424**. The neck **424** can extend between the head **420** and the anchor **422**.

(65) The nail anchor **422** and neck **424** can extend along an implant axis **405**. The nail head **420** can extend distally away from the neck **424** at an angle α . The nail head **420** can extend between a head first end **426** and a head second end **428** along a head first axis **407**. The angle α between the neck lateral surface and the head lateral surface can be approximately 25° . In other implementations of the invention, angle α may be in the range of 0° to 90° . Desirably, the angle α may be in the range of 0° to 60° . Desirably, the angle α may be in the range of 15° to 60° . In at least the embodiment depicted, the nail **402** and assembled implant **400** are bilaterally symmetrical with respect to the implant axis **405**, and with respect to the head first axis **407**.

(66) The nail head **420** includes a first opening **430a**. The first opening **430a** can be centered on the nail head **420**. The first opening **430a** can extend along a head second axis **410a**. The head second axis **410a** can be at an angle A1 to the implant axis **405**. The angle A1 can be between approximately 0° and 135° . Desirably, the angle A1 may be in the range of 30° to 60° . The head second axis **410a** may extend generally medially-laterally (ML) upon implantation. The first opening **430a** can extend between a head lateral side **415** and a head medial side **413**. The first opening **430a** can include threads **432a** for engagement with the first fastener **404a**. Other implementations may lack threads. A concave lip **434a** can encircle the first opening **430**. The head second axis **410a** can be at an angle γ with the head first axis **407**. Angle γ can be approximately 90° . Desirably, the angle γ may be in the range of 45° to 135° .

(67) The nail **402** can include a second opening **430b**. The second opening **430b** can be on the nail head **420** and/or the neck **424**. The second opening **430b** can be centered on a head third axis **410b**. The second opening **430b** can extend between the head lateral side **415** and the head medial side **413**. The head third axis **410b** can be at an angle A2 to the implant axis **405**. The angle A2 can be less than approximately 45° . Desirably, the angle A2 may be in the range of 0° to 90° . Desirably, the angle A2 may be in the range of 30° to 60° . The second opening **430b** can include concave lip **434b**. In some implementations, the second opening **430b** can include threads (not shown) for engagement with the second fastener **404b**.

(68) The head second end **428** can be wider than the neck **424** with respect to the superior-inferior dimension. The head second end **428** can include a first shoulder **440** and a second shoulder **442**. The shoulders **440**, **442** can project superiorly and inferiorly, respectively, away from the neck **424** at the intersection of the neck **424** and the head **420**. The first shoulder **440** can include a first proximal shoulder surface **441**. The second shoulder **442** can include a second proximal shoulder surface **443**. The proximal shoulder surfaces **441**, **443** can face proximally away from the head **420**. The proximal shoulder surfaces **441**, **443** can be at right angles to the neck superior and inferior sides **414**, **416**.

(69) The head **420** can include a transverse bore **448**. The transverse bore **448** can extend along a head fourth axis **411** which is perpendicular to the head first axis **407** and the head second axis **410a**. The head fourth axis **411** can extend generally superiorly-inferiorly (SI) upon implantation. The thickness of the head **420** between the medial **413** and lateral **415** sides can increase between the head first end **426** and the head second end **428** so that the thickest part of the head **420** is at the shoulders **440**, **442**. A greater thickness of the head **420** between the medial **413** and lateral **415** sides can shift the head **420** further outwardly with respect to a medial axis of the proximal metatarsal portion **2a**. The transverse bore **448** can extend through the thickest part of the head **420**.

(70) The neck **424** can extend between and connect the head **420** with the anchor **422**. The thickness of the neck **424** between the medial and lateral sides **410**, **412** can vary depending on the desired degree of shift of the metatarsal. The neck thickness can taper between the head **420** and the anchor **422**. The width of the neck **424** between the superior **414** and inferior **416** sides may also vary. The length of the nail **402** between the first and second ends **406**, **408** can vary, as can the

relative lengths of the head, neck, and/or anchor portions. The anchor **422** can be coaxial with the neck **424**. The anchor **422** can extend from the neck **424** to the second end **408** of the nail **402**. Both the thickness of the anchor **422** between the medial and lateral sides **410**, **412**, and the width of the anchor **422** between the superior **414** and inferior **416** sides can taper towards the nail second end **408**. This can promote easy insertion of the nail **402** into metatarsal 2.

(71) The anchor **422** at its second end **408** can be rounded, pointed, flattened, serrated, or another shape. The anchor **422** can include a plurality of bone engagement features **444** which may be shaped as teeth, scallops, serrations, or other shapes to promote engagement within bone. The neck **424** and anchor **422** can be free from or include openings for supplementary fixation or instrument connection.

(72) Fastener **404a** can include a fastener head **450a**, driving feature **459a**, fastener shaft **452a**, tip **454a**, threads **456a** for locking engagement with threads in the nail head **420** (e.g., opening **430a**), and/or threads **458a** for engagement in bone. Fastener **404b** can include a fastener head **450b**, driving feature **459b**, fastener shaft **452b**, tip **454b** and/or threads **458b** for engagement in bone. The fasteners **404a/404b** can be locking screw type fasteners; in other implementations the fasteners **404a/404b** can be locking or non-locking, and may be polyaxially adjustable or non-polyaxially adjustable. The nail **402** and fasteners **404a/404b** may comprise titanium, stainless steel, PEEK, nitinol, and/or other rigid biocompatible materials or combinations thereof.

(73) The implant system **400** can be used in place of the implant system **100** in the insertion process into a patient's foot as shown and described in FIGS. 1-20 and the accompanying text. The implant system **400** can require the following further steps to anchor the nail **402** into the prepared proximal metatarsal portion **2a**.

(74) An implant inserter such as the inserter **240** may be employed to implant the nail **402** into the proximal metatarsal portion **2a**. The implant engagement end **246** can cooperate with the openings **430a** and/or **430b** or otherwise with the nail **402**. The implant engagement end **246** can include threads **248** that can engage with nail threads **432a** (or nail threads in the second opening **430b**) to removably attach the nail **402** to the inserter **240**. The inserter **240** is moved to insert the anchor **422** and neck **424** proximally into the distal-facing surface **2c** and into the intramedullary canal of the metatarsal 2. Desirably, the engagement end **246** can cooperate with the opening **430b**, which can more closely align the inserter **240** with the anchor **422** and neck **424**. Accordingly, the inserter **240** can be used to apply force more directly into the proximal metatarsal portion **2a** as the anchor **422** and neck are inserted.

(75) The head **420** can be left distal to the proximal metatarsal portion **2a**. If needed, the inserter **240** may be tapped to drive the nail **402** into position in the proximal metatarsal portion **2a**. The nail **402** can be positioned so that the proximal surfaces **441**, **443** of shoulders **440**, **442** abut the distal-facing surface **2c** of the metatarsal. The nail head lateral side **415** can be immediately adjacent to medial surface **2d** of the distal metatarsal portion **2b**. With the nail **402** seated in the desired location, the inserter **240** can be rotated to disengage it from the implanted nail **402**.

(76) The fastener **404b** can fasten the nail **402** with the proximal metatarsal portion **2a**. A driver (e.g., screwdriver) can be employed to implant the fastener **404b**. The driver can couple with the driving feature **459b**. The tip **454b** and shaft **452b** can be inserted into the second opening **430b**. The fastener **404b** can be rotated to engage the threads **458b** with the proximal metatarsal portion **2a**. The installation of the fastener **404b** can advantageously secure the position of the nail **402** relative to the metatarsal 2. Accordingly, the suturing steps and alignment of the distal metatarsal portion **2b** relative to the nail **402** as shown in FIGS. 11A-12 and described in the accompanying description can be performed more accurately.

(77) Referring to FIGS. 26-29, a bunion correction implant system **500** according to another embodiment of the invention includes a nail **502**. The implant system **500** can be used in place of the implant system **400** in the insertion process into a patient's foot as shown and described above. The nail **502** may be implanted to extend longitudinally into the proximal metatarsal portion **2a** of

the resected metatarsal 2. The nail 502 can be fixed with fasteners in the manner similar to the fasteners 404a, 404b described above.

(78) The implant nail 502 can be a monolithic body extending from a first end 506 which may be a distal end, to a second end 508 which may be a proximal end. The nail 502 can have a medial side 510 which may be an outer side, a lateral side 512 which may be an inner side, a superior side 514, and/or an inferior side 516. The nail 502 can include a head 520. The head 520 can be one the first end 506. The nail 502 can include an anchor 522. The anchor 522 can be on the second end 508. The nail 502 can include a neck 524. The neck 524 can extend between the head 520 and the anchor 522.

(79) The nail anchor 522 and neck 524 can extend along an implant axis 505. The nail head 520 can extend distally away from the neck 524 at an angle α_1 along head axis 507. The angle α_1 between the neck lateral surface (along axis 505) and the head lateral surface (along head axis 507) can be approximately 25°. In other implementations of the invention, angle α_1 may be in the range of 0° to 90°. Desirably, the angle α_1 may be in the range of 0° to 60°. Desirably, the angle α_1 may be in the range of 15° to 60°. In at least the embodiment depicted, the nail 502 and assembled implant 500 are bilaterally symmetrical with respect to the implant axis 505.

(80) The nail head 520 includes a first opening 530a and a second opening 530b. The first and second head openings 530a, 530b can extend along respective first and second axes 510a, 510b. The first axis 510a can be at an angle B1 to the implant axis 505. The second axis 510b can be at an angle B2 to the implant axis 505. Angle B1 and B2 can be equal, although this is not required. The angles B1, B2 can be between approximately 0° and 135°. Desirably, the angles B1, B2 may be in the range of 30° to 90°. The axes 510a, 510b may extend generally medially-laterally (ML) upon implantation. The first and second openings 530a, 530b can extend between a head lateral side and a head medial side. The first and second openings 530a, 530b can include threads 532a, 532b for engagement with fasteners. Other implementations may lack threads. A concave lip can encircle the respective first and second openings 530a, 530b. The first and/or second axes 510a, 510b can be at perpendicular to the axis 507.

(81) The nail 502 can include a third opening 530c. The third opening 530c can be on the nail head 520 and/or the neck 522. The third opening 530c can be centered on a third axis 510c. The third opening 530c can extend between the head lateral side and the head medial side. The head third axis 510c can be at an angle B3 to the implant axis 505. The angle B3 can be less than approximately 45°. Desirably, the angle B3 may be in the range of 0° to 90°. Desirably, the angle B3 may be in the range of 30° to 60°. The third opening 530c can include a concave or tapered up lip. In some implementations, the third opening 530c can include threads (not shown) for engagement with a fastener.

(82) The neck 524 can extend between and connect the head 520 with the anchor 522. The neck 524 and/or anchor 522 may be generally rectangular in cross-section. The anchor 522 can extend from the neck 524 to the second end 508 of the nail 502. Both the thickness of the anchor 522 between the medial and lateral sides 510, 512, and the width of the anchor 522 between the superior 514 and inferior 516 sides can taper towards the nail second end 508. This can promote easy insertion of the nail 502 into metatarsal 2. The head 520 can include a transverse bore 548. The transverse bore 548 can extend generally superiorly-inferiorly (SI) upon implantation.

(83) The anchor 522 at its second end 508 can be rounded, pointed, flattened, serrated, or another shape. The anchor 522 can include a plurality of bone engagement features 544 which may be shaped as teeth, scallops, serrations, or other shapes to promote engagement within bone. The neck 524 and anchor 522 can be free from or include openings for supplementary fixation or instrument connection.

(84) Any methods disclosed herein comprise one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the

embodiment, the order and/or use of specific steps and/or actions may be modified.

(85) Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

(86) Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, Figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.

(87) Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the invention.

(88) While specific embodiments and applications of the present invention have been illustrated and described, it is to be understood that the invention is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present invention disclosed herein without departing from the spirit and scope of the invention.

(89) The terms “approximately,” “about,” and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, in some embodiments, as the context may dictate, the terms “approximately,” “about,” and “substantially,” may refer to an amount that is within less than or equal to 10% of the stated amount. The term “generally” as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. As an example, in certain embodiments, as the context may dictate, the term “generally parallel” can refer to something that departs from exactly parallel by less than or equal to 20 degrees. Given ranges are inclusive of endpoints.

Claims

1. An implant nail for correcting a bunion formed at a joint between a metatarsal and a great toe, the implant nail comprising: a monolithic body comprising: a head at a first end of the monolithic body, the head having a bone-facing surface and an outer surface opposite the bone-facing surface, the head comprising: a head first end including a shoulder; a head second end; a transverse bore extending along a superior-inferior axis of the head, the transverse bore disposed closer to the bone-facing surface than the outer surface; and a first bone fixation hole extending through the head, the first bone fixation hole oriented along a first hole axis; an anchor at a second end of the monolithic body; and a neck disposed between the second head end and the anchor, the neck comprising a second bone fixation hole extending through the neck, the second bone fixation hole oriented along a second hole axis, the second hole axis being off-axis from the first hole axis.
2. The implant nail of claim 1, wherein an implant axis of the anchor forms a first angle with a head axis of the head, the first angle being less than 90°.

3. The implant nail of claim 2, wherein the implant axis forms a second angle with the second hole axis, the second angle being between 30° and 60.
 4. The implant nail of claim 3, wherein the first angle between the implant axis and the head axis is in the range of 15° to 60°.
 5. The implant nail of claim 3, wherein the first hole axis is perpendicular to a plane defined by a lateral side of the head.
 6. The implant nail of claim 3, wherein the first and second angles are in a plane parallel with a superior side of the nail.
 7. The implant nail of claim 3, wherein the shoulder includes superior and inferior portions extending at right angles to respective superior and inferior sides of the neck.
 8. The implant nail of claim 3, wherein the second bone fixation hole is configured to receive a second fastener such that a head of the second fastener is fully recessed within the neck.
 9. The implant nail of claim 3, wherein the first and second angles are measured in a plane parallel with a superior view of the implant.
 10. The implant nail of claim 1, wherein the first bone fixation hole diverges away from the second bone fixation hole as the first bone fixation hole extends from the outer surface to the bone-facing surface.
 11. The implant nail of claim 10, wherein the transverse bore is disposed between the first and second bone fixation holes.
 12. The implant nail of claim 11, wherein the first and second bone fixation holes intersect each other at the outer surface.
 13. The implant nail of claim 1, wherein the first and second bone fixation holes are aligned with each other along a length of the implant nail.
 14. The implant nail of claim 13, wherein the first and second bone fixation holes are the only bone fixation holes of the implant nail.
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